

Empathy-Driven Virtual Reality for ADHD: Integrating Emotional Support into Child-Computer Interaction

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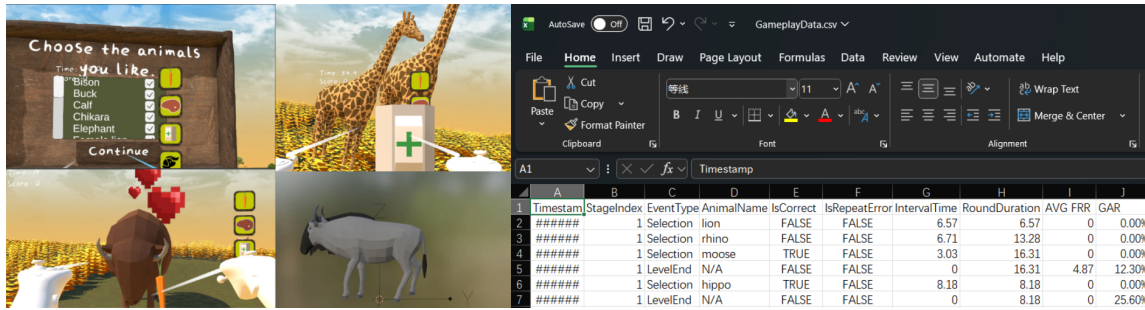


Figure 1: System Overview and Data Logging Architecture. **Left & Middle:** The VR user experience, featuring the **Companion Selection Interface** (Top-Left), the **Rescue Mission** with injured giraffes (Top-Middle), and **Caretaking Interactions** (Bottom-Left). **Right:** The backend CSV data log used for evaluation. Key columns include: EventType (logs actions like 'Selection' or 'LevelEnd'), IsCorrect (tracks accuracy), IntervalTime (measures latency between attempts, used to calculate FRR), and GAR (records the Gaze Adherence Ratio for attention tracking).

ABSTRACT

Traditional Attention Deficit/Hyperactivity Disorder (ADHD) therapies often lack engagement, and existing Virtual Reality (VR) systems frequently overlook empathetic interaction. To address this, a Unity-based VR prototype incorporating empathetic design is introduced. The system features avatars that provide encouragement and emotional support to sustain child motivation. Preliminary testing has verified technical feasibility. Furthermore, an evaluation framework is proposed to assess impacts on attention and emotional regulation. By integrating empathy into VR interventions, this study contributes to the development of child-centered digital therapeutics, which is designed to enhance both effectiveness and user acceptance.

Index Terms: Virtual Reality, Attention Deficit/Hyperactivity Disorder (ADHD), Empathetic Interaction, Child-Computer Interaction.

1 INTRODUCTION

Attention-Deficit/Hyperactivity Disorder (ADHD) is characterized by persistent patterns of inattention and hyperactivity that interfere with development [1]. While Virtual Reality (VR) offers immersive environments that can optimize sensory load for intervention [2, 3], many systems still emphasize cognitive drills and performance, leaving opportunities for explicit emotional safety and motivational support. Standard binary feedback (correct/incorrect) can trigger frustration and disengagement in neurodiverse children [4].

This study introduces a VR prototype that treats *empathy as a first-class design constraint*. Unlike “performance-first” systems, this “empathy-first” design operationalizes empathy as a feedback

policy that acknowledges effort, normalizes mistakes, and encourages persistence, aligned with prior work on empathy in interactive agents and therapeutic VR contexts [5]. The objective is to show that emotionally supportive feedback improves time-on-task and reduces error-recovery latency compared to neutral feedback.

2 SYSTEM DESIGN AND IMPLEMENTATION

The prototype is developed using Unity and XR technologies for the PICO 4 standalone headset. The design philosophy centers on an *Empathy Manager*, a modular system architecture designed to monitor user behavior and regulate emotional feedback to foster psychological safety.

Data-Driven Empathy Framework. The system uses a real-time data pipeline that ingests sensor inputs (e.g., head tracking, gaze vectors) and interaction metrics (e.g., error frequency, response time) into a central Empathy Engine. An event-driven design decouples behavioral analysis from gameplay so empathetic responses (supportive NPC behaviors and audio-visual cues) can be triggered without disrupting task flow. Eye-tracking-informed attention signals align with digital biomarkers used for ADHD assessment and monitoring [6].

Empathy Engine (Interpretable Rule-Based Baseline). To address concerns about overstating complexity, the empathy regulation in our prototype is intentionally implemented as a lightweight, interpretable rule set integrated into the runtime manager scripts. The engine maps task events (correct/incorrect actions and periods of inaction) to NPC bubble prompts and reinforcement effects.

Signals. Two interaction signals are tracked: (i) *ErrorsInLastMinute*, the number of incorrect actions within a rolling window of $W=60s$, and (ii) *TimeSinceLastAction*, the elapsed time since the last intentional input (e.g., selection, confirmation, or task-relevant manipulation). The baseline struggle predicate used by the system is: $IsUserStruggling = (ErrorsInLastMinute > 3) \vee (TimeSinceLastAction > 20s)$.

Response Mapping (Equivalent to Two Threshold Rules). When $ErrorsInLastMinute > 3$, the system treats the user as frustrated and triggers a “Soothe” response, presented as a supportive

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NPC dialogue bubble (e.g., reassurance and brief hints) rather than punitive feedback. When $\text{TimeSinceLastAction} > 20\text{s}$, the system treats the user as off-task and triggers a gentle re-engagement prompt (NPC bubble cue). Outside these conditions, correct actions trigger short praise bubbles, and task completion triggers positive reinforcement (e.g., confetti VFX; Figure 2). The thresholds (3 errors/minute and 20s inaction) are configurable and were selected via iterative pilot tuning to balance sensitivity and intrusiveness; in future user studies, we will recalibrate these values based on empirical session logs and outcome analyses.

Sensory and Cognitive Safety. To mitigate sensory overload (a common barrier for children with ADHD), the system uses low-poly rounded assets with warm, desaturated palettes to reduce clutter/cognitive load, and applies automatic “ducking” so ambient sounds lower when the empathetic guide speaks, preserving intelligibility and reducing auditory fatigue.

Interaction Rhythm and Narrative Context. To mitigate impulsivity and filter out motor tremors, interaction uses a stabilized “Dwell-and-Select” mechanism: the child maintains gaze or controller focus on a target for a short duration to confirm intention. This approach aligns with design strategies for improving game learnability and control [7]. In the current prototype, the dwell duration is set to $\tau_{\text{dwell}} = 2\text{s}$ (configurable), pilot-tuned to reduce accidental activation while maintaining responsiveness.

The intervention is framed within a “Guardian Narrative.” The primary loop emphasizes **caretaking interactions (e.g., feeding companions)** and autonomous animal selection, leveraging playful learning principles [8]. To evaluate empathy-driven attention shifting, a “Rescue Mission” introduces a sporadic high-priority event: an *injured giraffe* interrupts the flow and requires inhibition and attention redirection to provide aid.

Interactive Empathic Avatars. To foster social connection, the virtual guide uses a Finite State Machine (FSM) with proximity and gaze detection via the XR Interaction Toolkit to maintain eye contact. The Animator Controller tracks parameters such as `IsUserStruggling`; repeated errors trigger a “Concerned” state that provides non-judgmental visual prompts to re-engage the user.

Backend Data Logging. To support quantitative evaluation, a robust `DataManager` module is implemented to capture high-frequency interaction events. As shown in Figure 1 (Right), the system serializes timestamped logs into local CSV files, including `EventType`, `IsCorrect`, and `IntervalTime`. Derived attention metrics such as GAR (Gaze Adherence Ratio) are also recorded for immediate analysis. Logging runs asynchronously and enforces data minimization to support ethical pediatric research.

Performance Optimization. Maintaining stable frame rate is critical to prevent cybersickness. We use Object Pooling for dynamic assets (e.g., animals, confetti effects) to avoid spikes from frequent `Instantiate/Destroy`, and configure URP with baked lightmaps to reduce real-time rendering overhead.

Autonomy via Companion Selection. To foster agency, a “Companion Selection Interface” (Figure 1, Top-Left) lets children choose which animals appear. Safety is enforced via boundary testing: a failsafe prevents “Select None”, and deselected animals are excluded from the object pool to avoid accidental exposure to phobia triggers.

Multi-Tiered Feedback Loop. The system moves beyond simple gamification to a feedback model designed to support motivation [4]: *Constructive guidance* during struggle states (Figure 2, Left) avoids negative reinforcement via encouraging cues, while *positive reinforcement* on completion (Figure 2, Right) uses celebration effects (e.g., confetti) to validate achievements.

3 PROPOSED EVALUATION

A future controlled evaluation is proposed to contrast the “Empathy-First” (EF) condition against a “Neutral-Feedback” (NF)



Figure 2: The Multi-Tiered Empathetic Feedback System. Left: Constructive guidance (“Don’t worry...”) during a struggle state. Right: Positive reinforcement (confetti) upon successful task completion.

control. The study will utilize two novel quantitative metrics defined in this work and attention-relevant eye-tracking signals used in ADHD digital biomarker research [6].

Frustration Recovery Rate (FRR). Defined as the time interval Δt required for a user to re-attempt a task after a failure event; EF is hypothesized to yield a lower FRR (higher resilience).

$$FRR = t_{\text{next_input}} - t_{\text{error_event}} \quad (1)$$

where $t_{\text{next_input}}$ denotes the timestamp of the immediate next interaction attempt, and $t_{\text{error_event}}$ represents the timestamp when the initial failure occurred.

Gaze Adherence Ratio (GAR). Utilizing eye-tracking capabilities, GAR measures the proportion of time the user’s gaze intersects with the task-relevant Region of Interest (ROI):

$$GAR = \left(\frac{\sum_{i=1}^n \mathbb{I}(V_g^{(i)} \in ROI)}{T_{\text{total}}} \right) \times 100\% \quad (2)$$

Here, n is the total number of frames recorded; $\mathbb{I}$ is the indicator function (returning 1 if the condition is met, 0 otherwise); $V_g^{(i)}$ represents the gaze vector at frame i ; and T_{total} is the total duration of the session. Higher GAR values in the EF condition would confirm that empathetic companions help anchor visual attention better than neutral distractors.

4 CONCLUSION AND FUTURE WORK

This work contributes to Child-Computer Interaction (CCI) by proposing a shift from “correction” to “connection”. By integrating empathy as a technical primitive—through interpretable struggle detection and layered feedback logic—the system aims to destigmatize failure in ADHD interventions. Future work will focus on clinical validation of FRR and GAR, including threshold calibration and personalization analysis.

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